Wine Quality EDA

Red Wine

Summary Table

	count	mean	std	variance
fixed acidity	1599.0	8.319637	1.741096	3.031416
volatile acidity	1599.0	0.527821	0.179060	0.032062
citric acid	1599.0	0.270976	0.194801	0.037947
residual sugar	1599.0	2.538806	1.409928	1.987897
chlorides	1599.0	0.087467	0.047065	0.002215
free sulfur dioxide	1599.0	15.874922	10.460157	109.414884
total sulfur dioxide	1599.0	46.467792	32.895324	1082.102373
density	1599.0	0.996747	0.001887	0.000004
pH	1599.0	3.311113	0.154386	0.023835
sulphates	1599.0	0.658149	0.169507	0.028733
alcohol	1599.0	10.422983	1.065668	1.135647

	min	q1	median	q3	max	skewness	kurtosis
fixed acidity	4.60000	7.1000	7.90000	9.200000	15.90000	0.982751	1.132143
volatile acidity	0.12000	0.3900	0.52000	0.640000	1.58000	0.671593	1.225542
citric acid	0.00000	0.0900	0.26000	0.420000	1.00000	0.318337	-0.788998
residual sugar	0.90000	1.9000	2.20000	2.600000	15.50000	4.540655	28.617595
chlorides	0.01200	0.0700	0.07900	0.090000	0.61100	5.680347	41.715787
free sulfur dioxide	1.00000	7.0000	14.00000	21.000000	72.00000	1.250567	2.023562
total sulfur dioxide	6.00000	22.0000	38.00000	62.000000	289.00000	1.515531	3.809824
density	0.99007	0.9956	0.99675	0.997835	1.00369	0.071288	0.934079
pH	2.74000	3.2100	3.31000	3.400000	4.01000	0.193683	0.806943
sulphates	0.33000	0.5500	0.62000	0.730000	2.00000	2.428672	11.720251
alcohol	8.40000	9.5000	10.20000	11.100000	14.90000	0.860829	0.200029

Distribution Plots

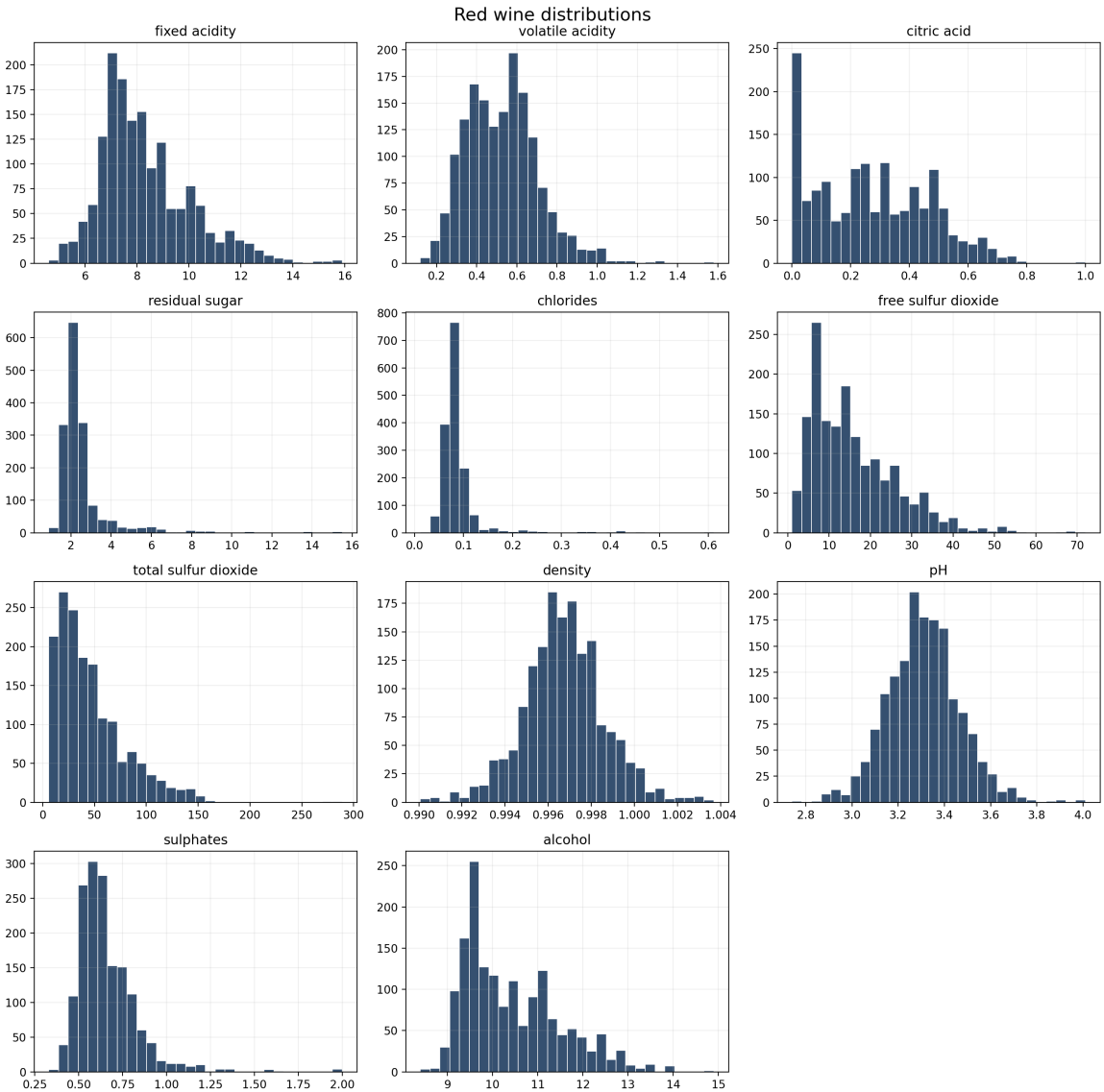

Figure 1: Red wine distributions for all 11 variables

Pearson Correlation

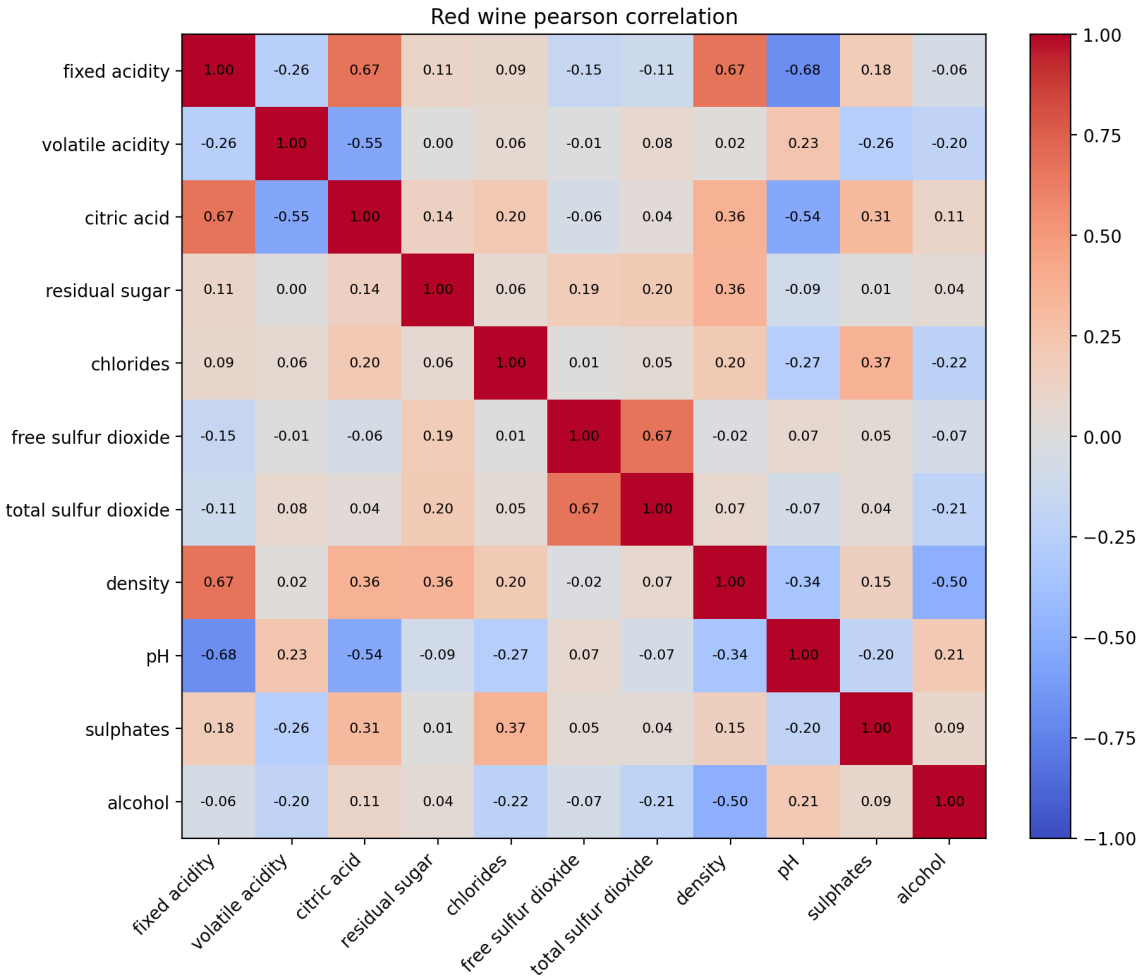

Figure 2: Red wine Pearson correlation matrix

Spearman Correlation

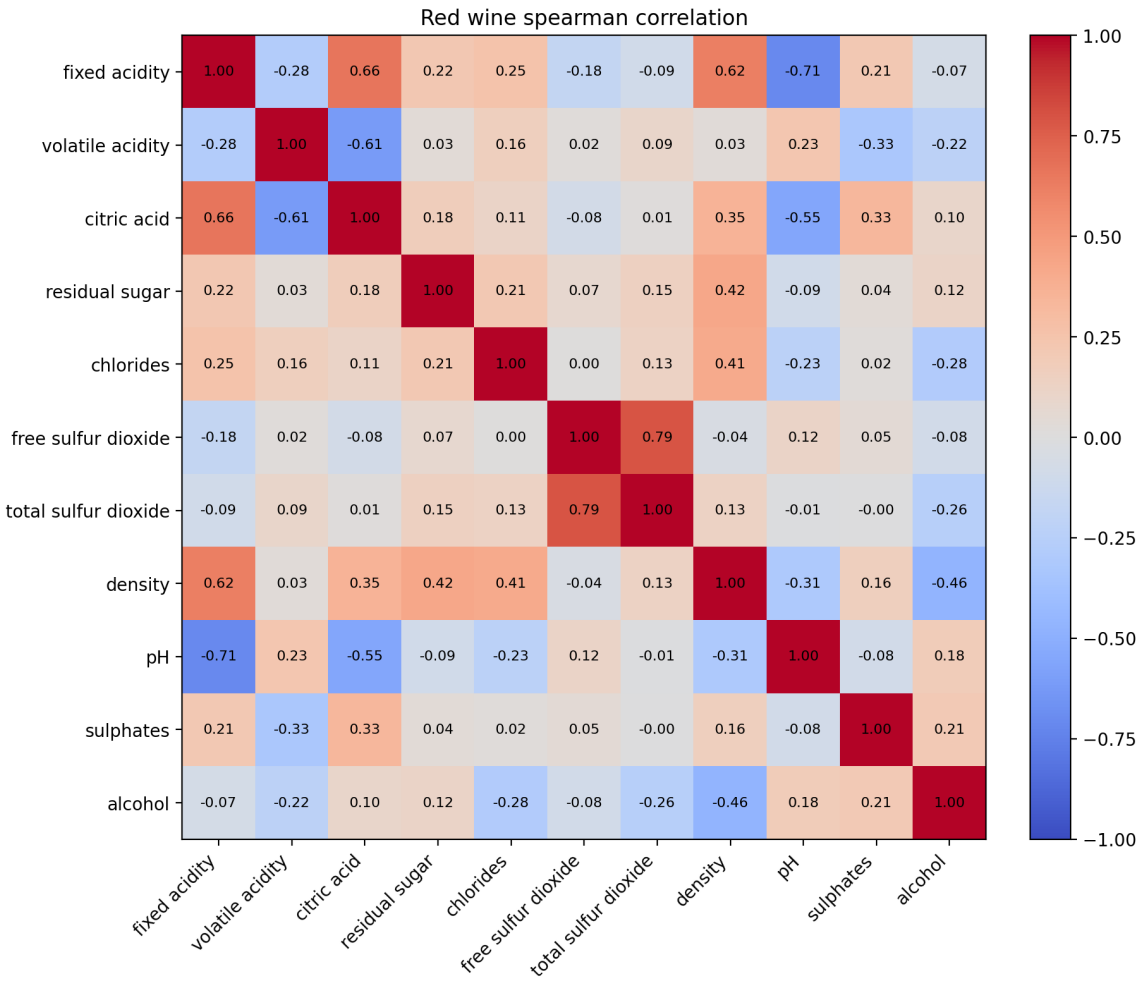

Figure 3: Red wine Spearman correlation matrix

PCA Summary Table

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
0	PC1	3.101072	0.281739	0.281739
1	PC2	1.927115	0.175083	0.456822
2	PC3	1.551514	0.140958	0.597781
3	PC4	1.213992	0.110294	0.708074
4	PC5	0.959892	0.087208	0.795283
5	PC6	0.660021	0.059964	0.855247
6	PC7	0.584157	0.053072	0.908319
7	PC8	0.423221	0.038451	0.946770

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
8	PC9	0.344858	0.031331	0.978101
9	PC10	0.181447	0.016485	0.994586
10	PC11	0.059596	0.005414	1.000000

PCA Eigenvalues

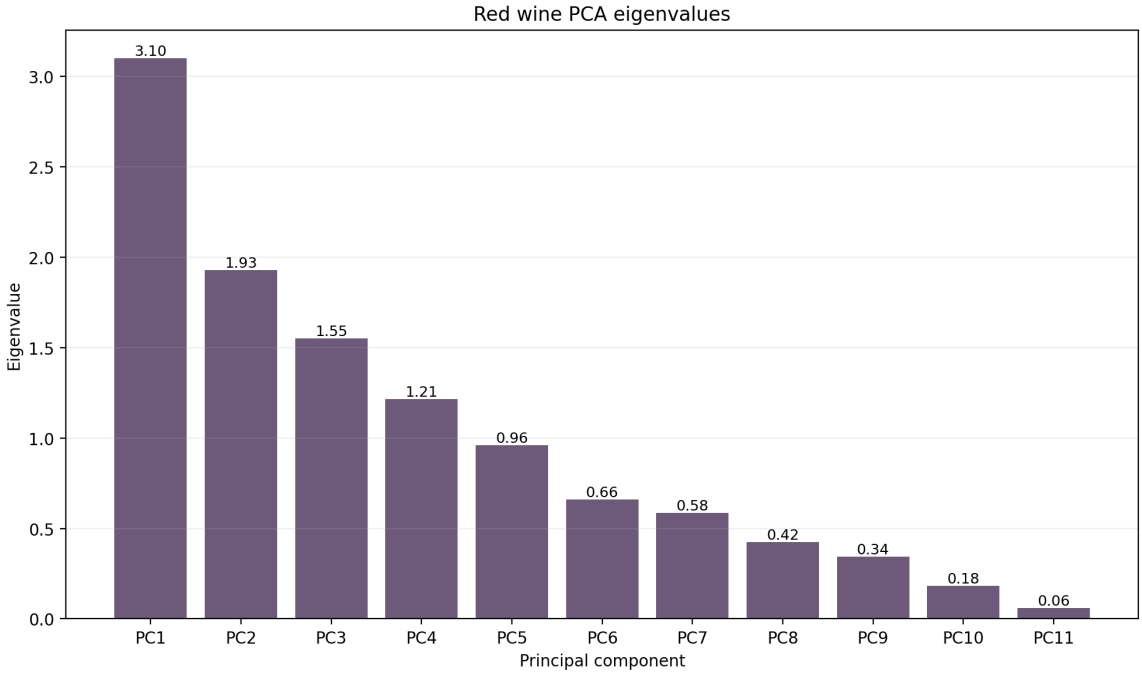

Figure 4: Red wine PCA eigenvalues

White Wine

Summary Table

	count	mean	std	variance
fixed acidity	4898.0	6.854788	0.843868	0.712114
volatile acidity	4898.0	0.278241	0.100795	0.010160
citric acid	4898.0	0.334192	0.121020	0.014646
residual sugar	4898.0	6.391415	5.072058	25.725770
chlorides	4898.0	0.045772	0.021848	0.000477
free sulfur dioxide	4898.0	35.308085	17.007137	289.242720
total sulfur dioxide	4898.0	138.360657	42.498065	1806.085491
density	4898.0	0.994027	0.002991	0.000009
pH	4898.0	3.188267	0.151001	0.022801
sulphates	4898.0	0.489847	0.114126	0.013025
alcohol	4898.0	10.514267	1.230621	1.514427

	min	q1	median	q3	max	skewness	kurtosis
fixed acidity	3.80000	6.300000	6.80000	7.3000	14.20000	0.647751	2.172178
volatile acidity	0.08000	0.210000	0.26000	0.3200	1.10000	1.576980	5.091626
citric acid	0.00000	0.270000	0.32000	0.3900	1.66000	1.281920	6.174901
residual sugar	0.60000	1.700000	5.20000	9.9000	65.80000	1.077094	3.469820
chlorides	0.00900	0.036000	0.04300	0.0500	0.34600	5.023331	37.564600
free sulfur dioxide	2.00000	23.000000	34.00000	46.0000	289.00000	1.406745	11.466342
total sulfur dioxide	9.00000	108.000000	134.00000	167.0000	440.00000	0.390710	0.571853
density	0.98711	0.991723	0.99374	0.9961	1.03898	0.977773	9.793807
pH	2.72000	3.090000	3.18000	3.2800	3.82000	0.457783	0.530775
sulphates	0.22000	0.410000	0.47000	0.5500	1.08000	0.977194	1.590930
alcohol	8.00000	9.500000	10.40000	11.4000	14.20000	0.487342	-0.698425

Distribution Plots

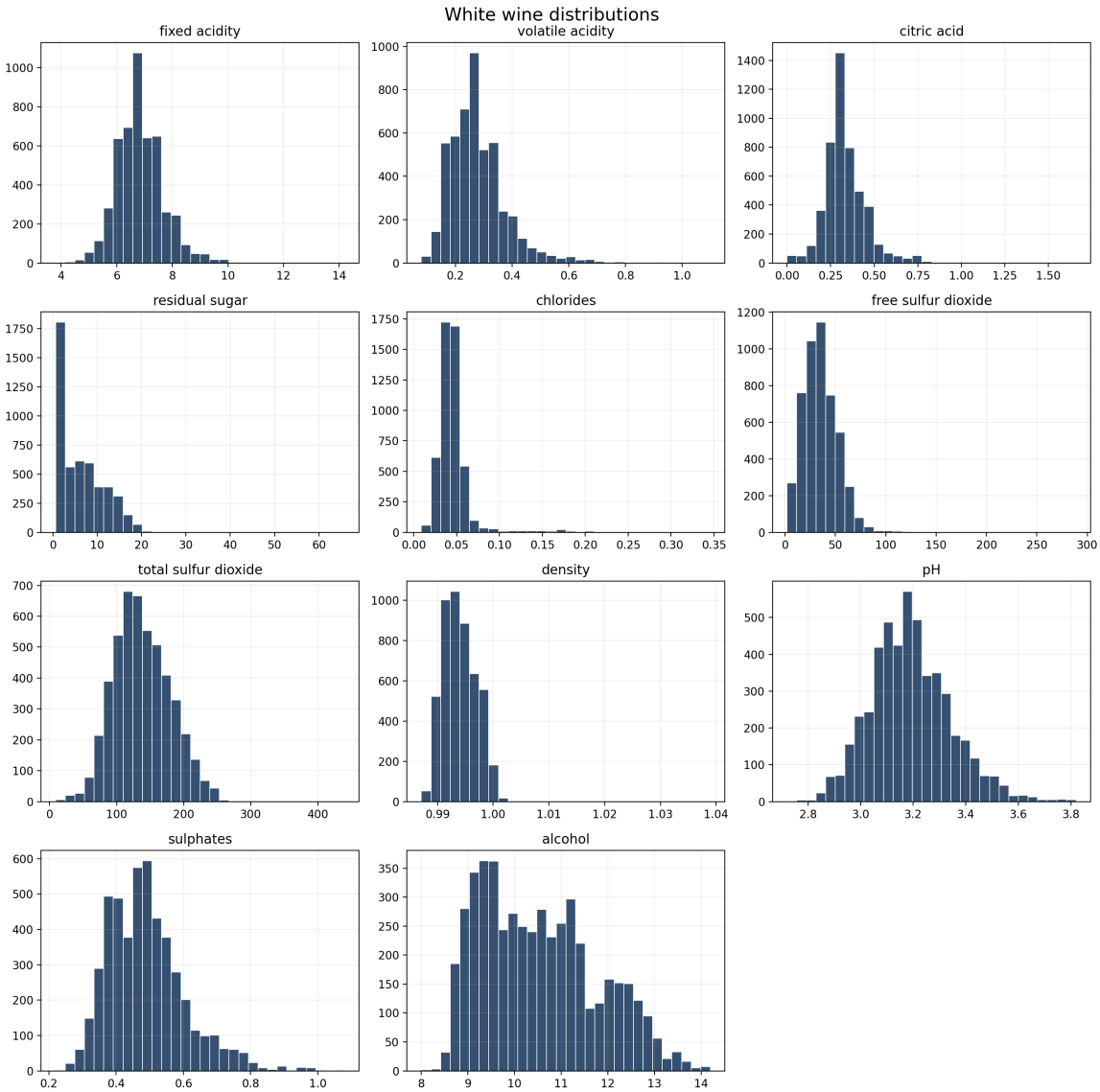

Figure 5: White wine distributions for all 11 variables

Pearson Correlation

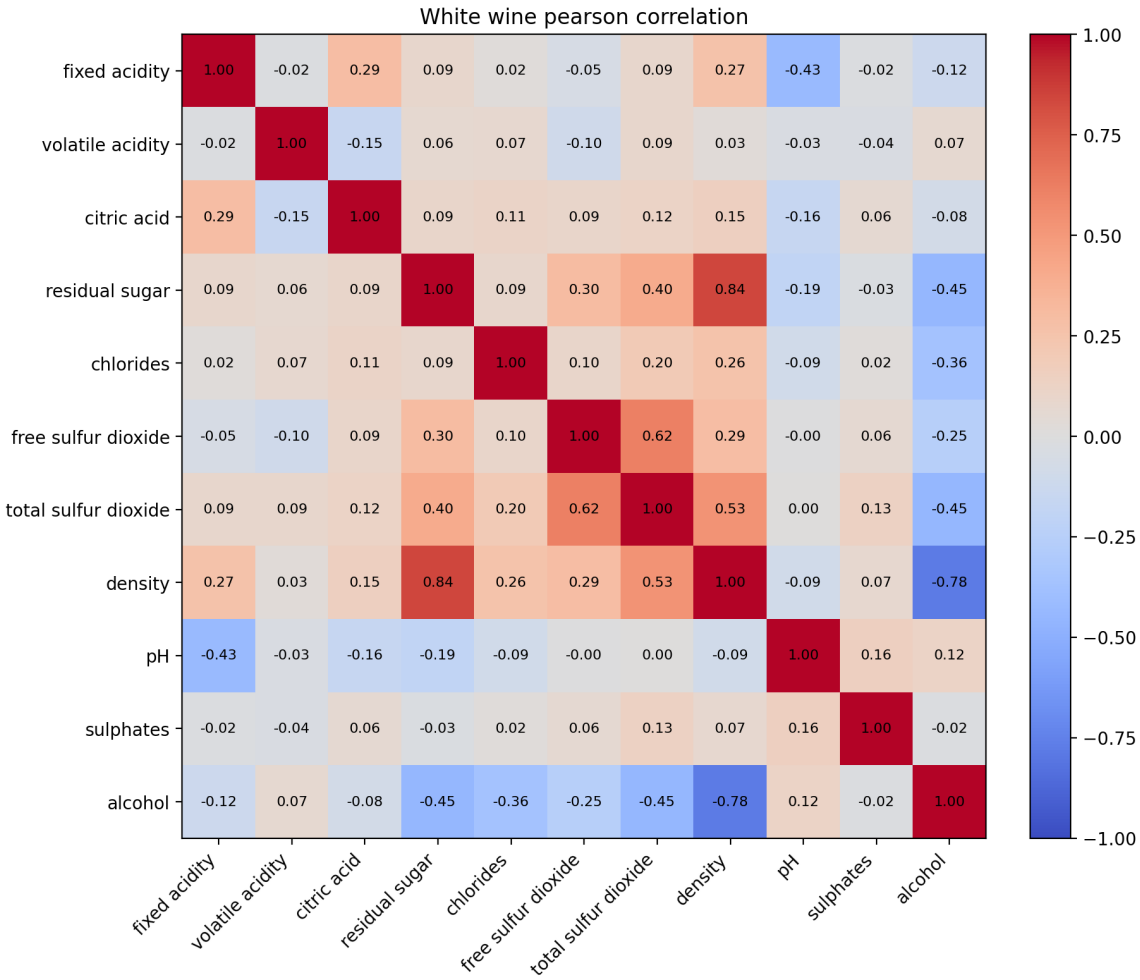

Figure 6: White wine Pearson correlation matrix

Spearman Correlation

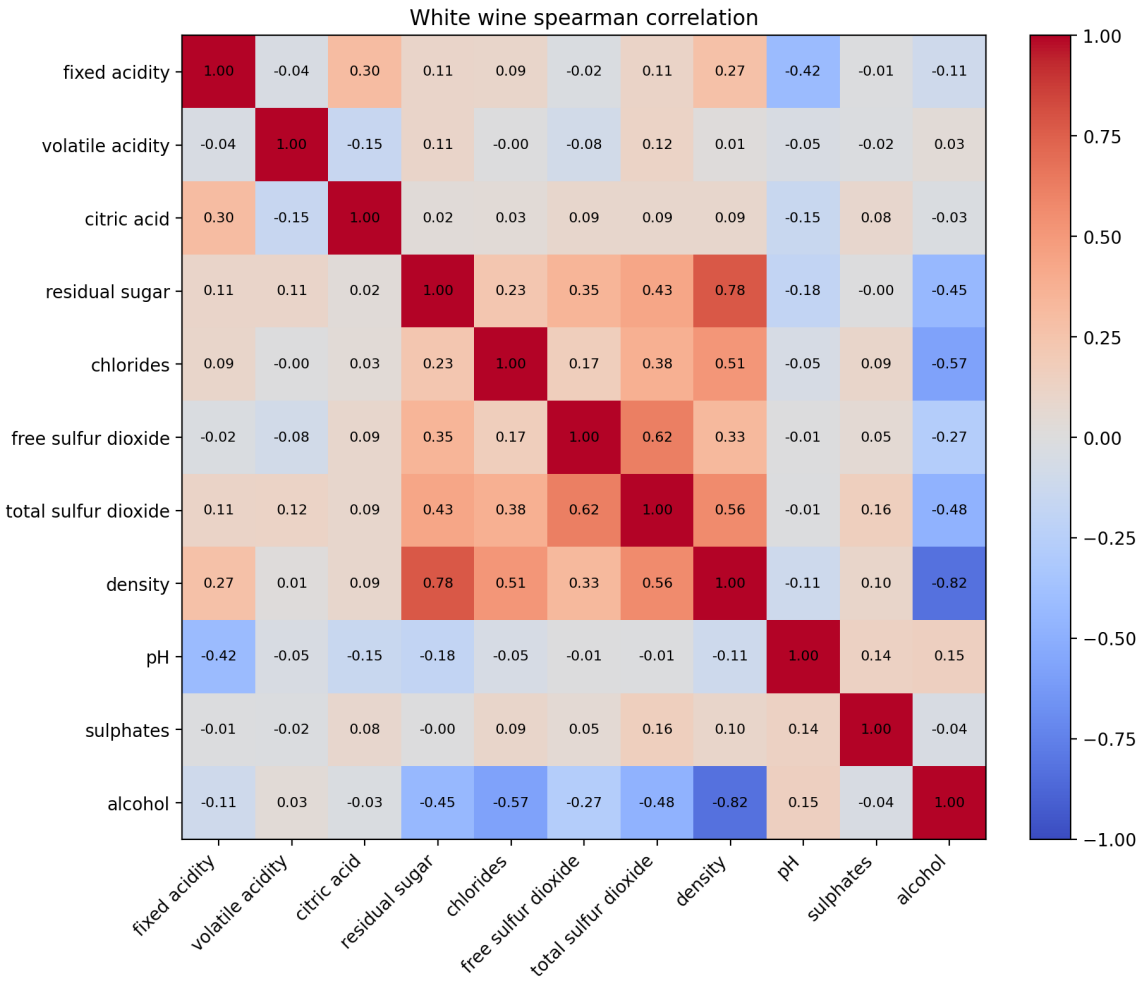

Figure 7: White wine Spearman correlation matrix

PCA Summary Table

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
0	PC1	3.222912	0.292932	0.292932
1	PC2	1.575562	0.143204	0.436136
2	PC3	1.221921	0.111061	0.547197
3	PC4	1.018730	0.092593	0.639790
4	PC5	0.973533	0.088485	0.728275
5	PC6	0.938933	0.085340	0.813615
6	PC7	0.726746	0.066054	0.879669
7	PC8	0.599481	0.054487	0.934156

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
8	PC9	0.414228	0.037649	0.971806
9	PC10	0.289546	0.026317	0.998123
10	PC11	0.020653	0.001877	1.000000

PCA Eigenvalues

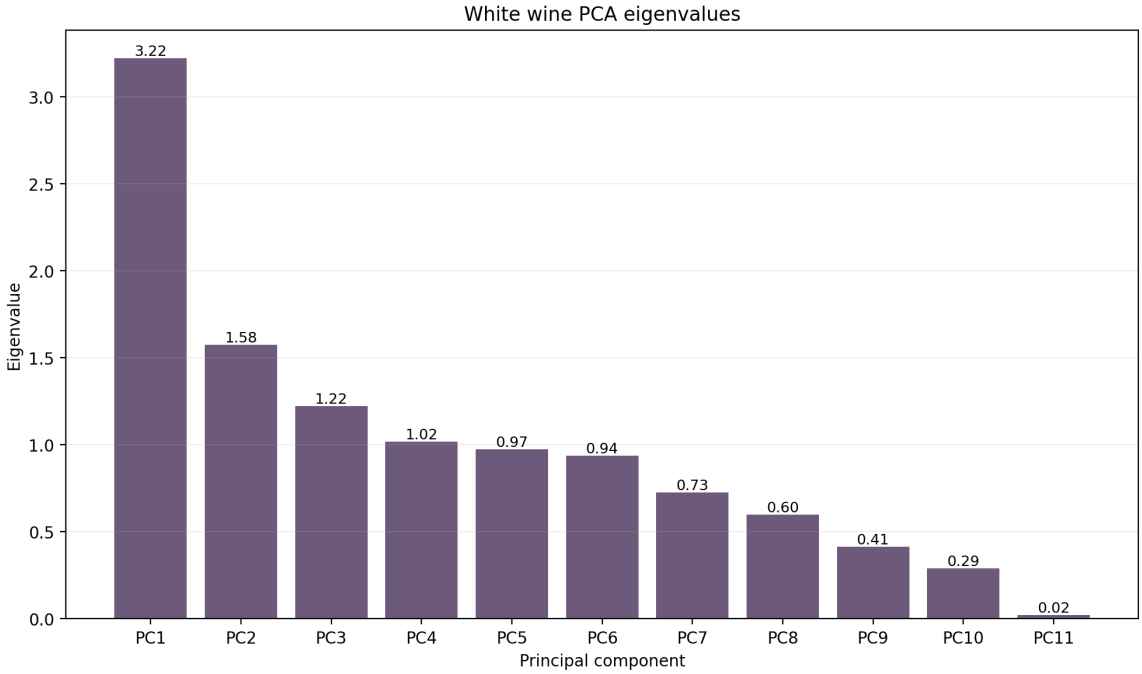

Figure 8: White wine PCA eigenvalues

Combined

Summary Table

	count	mean	std	variance
fixed acidity	6497.0	7.215307	1.296434	1.680740
volatile acidity	6497.0	0.339666	0.164636	0.027105
citric acid	6497.0	0.318633	0.145318	0.021117
residual sugar	6497.0	5.443235	4.757804	22.636696
chlorides	6497.0	0.056034	0.035034	0.001227
free sulfur dioxide	6497.0	30.525319	17.749400	315.041192
total sulfur dioxide	6497.0	115.744574	56.521855	3194.720039
density	6497.0	0.994697	0.002999	0.000009
pH	6497.0	3.218501	0.160787	0.025853
sulphates	6497.0	0.531268	0.148806	0.022143
alcohol	6497.0	10.491801	1.192712	1.422561

	min	q1	median	q3	max	skewness	kurtosis
fixed acidity	3.80000	6.40000	7.00000	7.70000	15.90000	1.723290	5.061161
volatile acidity	0.08000	0.23000	0.29000	0.40000	1.58000	1.495097	2.825372
citric acid	0.00000	0.25000	0.31000	0.39000	1.66000	0.471731	2.397239
residual sugar	0.60000	1.80000	3.00000	8.10000	65.80000	1.435404	4.359272
chlorides	0.00900	0.03800	0.04700	0.06500	0.61100	5.399828	50.898051
free sulfur dioxide	1.00000	17.00000	29.00000	41.00000	289.00000	1.220066	7.906238
total sulfur dioxide	6.00000	77.00000	118.00000	156.00000	440.00000	-0.001177	-0.371664
density	0.98711	0.99234	0.99489	0.99699	1.03898	0.503602	6.606067
pH	2.72000	3.11000	3.21000	3.32000	4.01000	0.386839	0.367657
sulphates	0.22000	0.43000	0.51000	0.60000	2.00000	1.797270	8.653699
alcohol	8.00000	9.50000	10.30000	11.30000	14.90000	0.565718	-0.531687

Distribution Plots

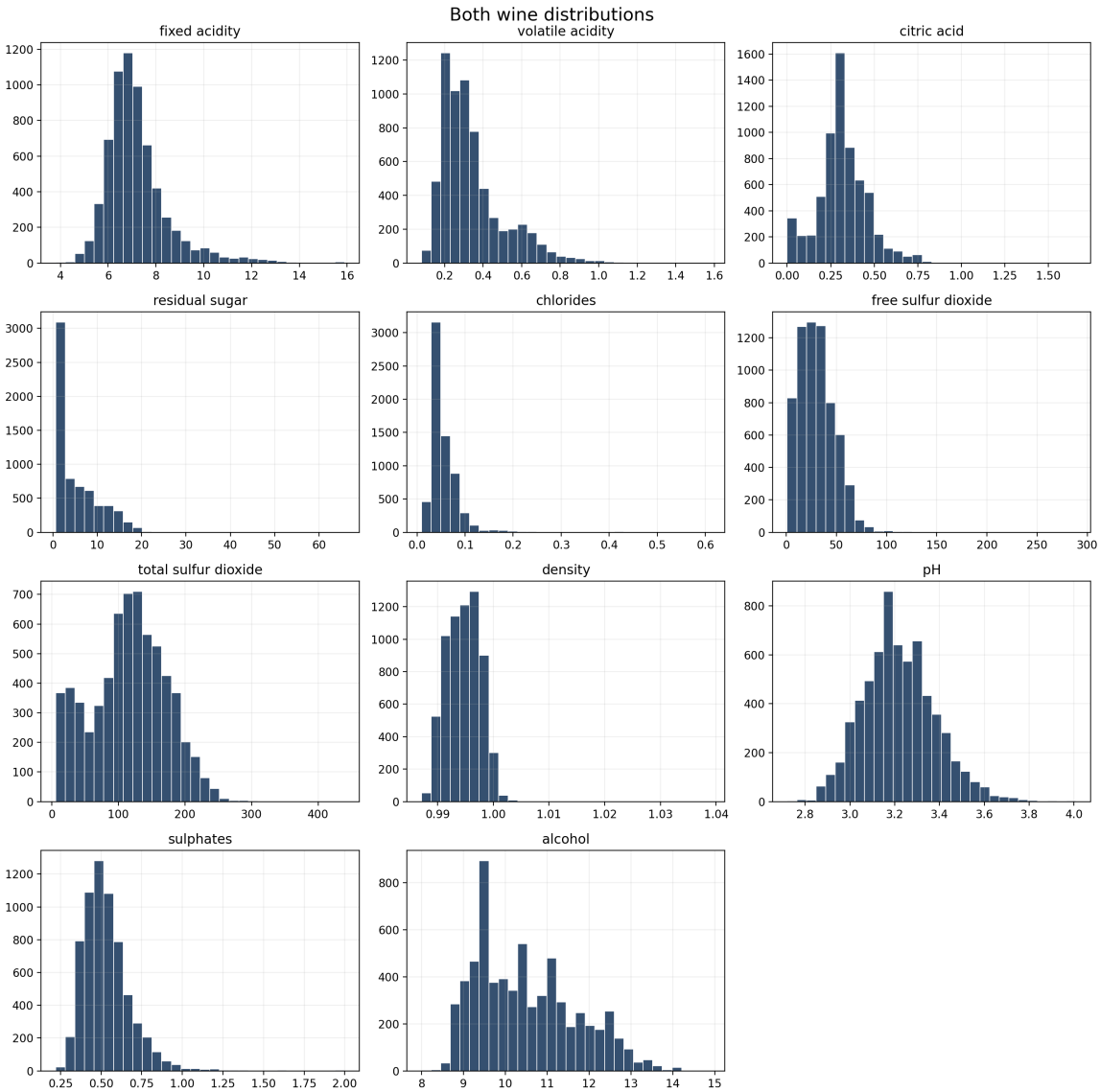

Figure 9: Combined wine distributions for all 11 variables

Pearson Correlation

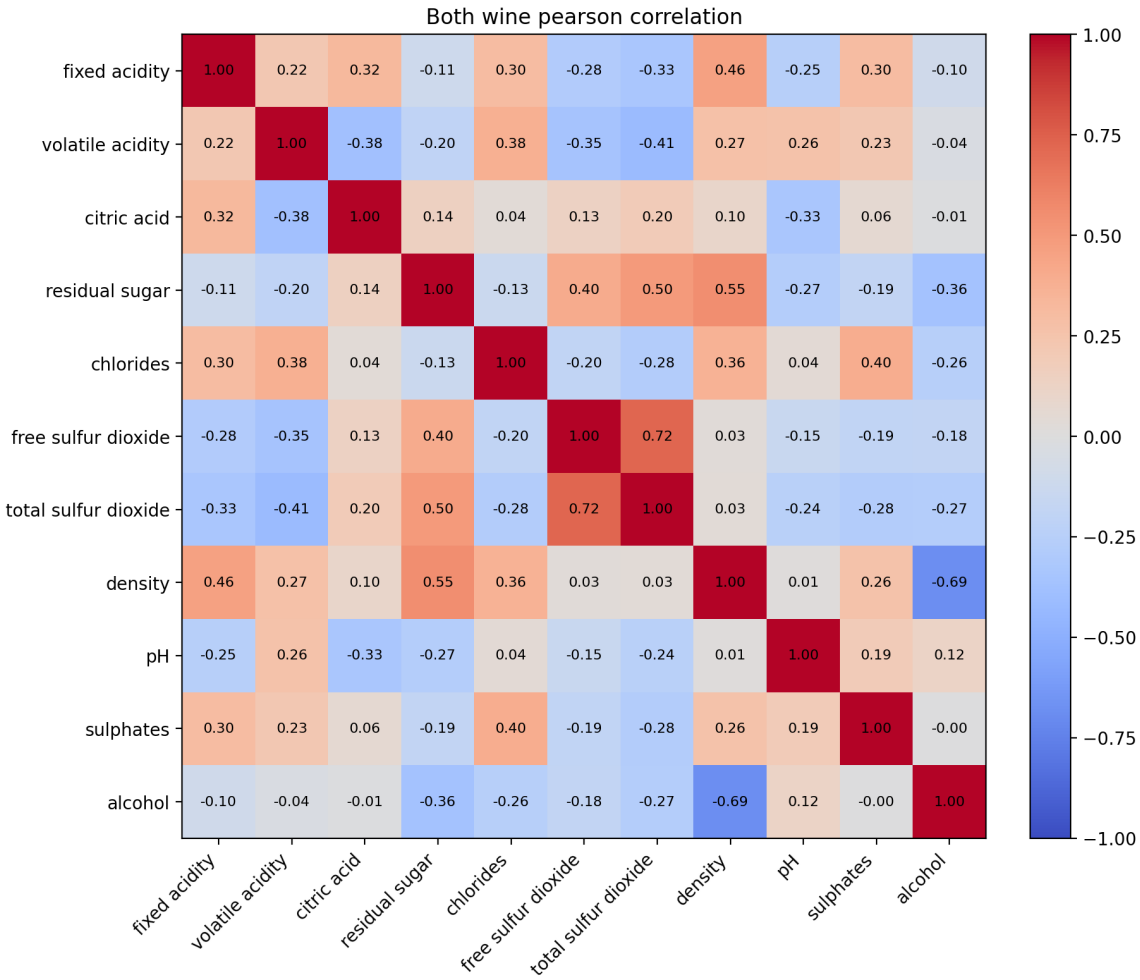

Figure 10: Combined wine Pearson correlation matrix

Spearman Correlation

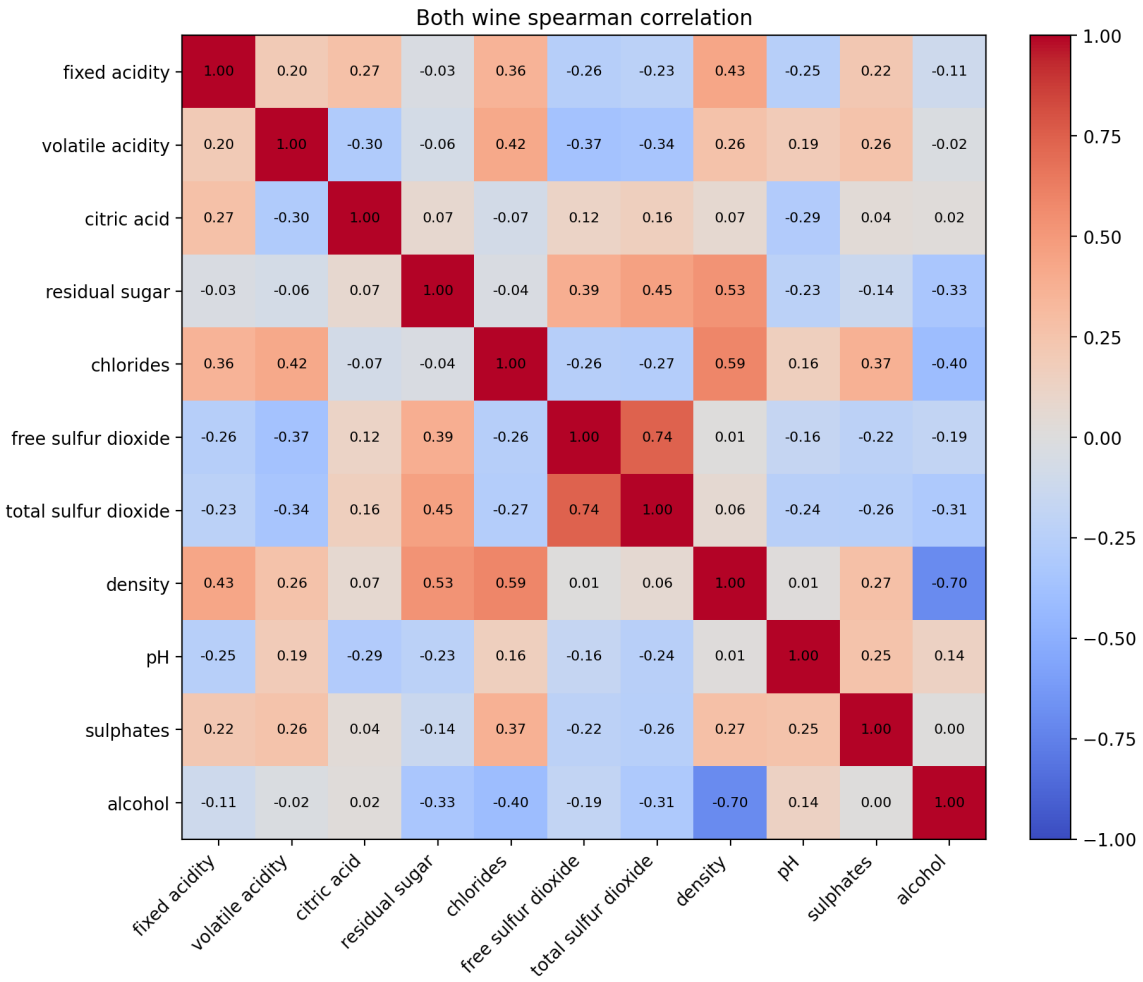

Figure 11: Combined wine Spearman correlation matrix

PCA Summary Table

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
0	PC1	3.030335	0.275443	0.275443
1	PC2	2.494210	0.226711	0.502154
2	PC3	1.556587	0.141486	0.643640
3	PC4	0.970701	0.088232	0.731872
4	PC5	0.719986	0.065443	0.797315
5	PC6	0.607405	0.055210	0.852525
6	PC7	0.523239	0.047560	0.900085
7	PC8	0.501587	0.045592	0.945677

	component	eigenvalue	explained_variance_ratio	cumulative_explained_variance_ratio
8	PC9	0.337076	0.030639	0.976316
9	PC10	0.227731	0.020700	0.997015
10	PC11	0.032836	0.002985	1.000000

PCA Eigenvalues

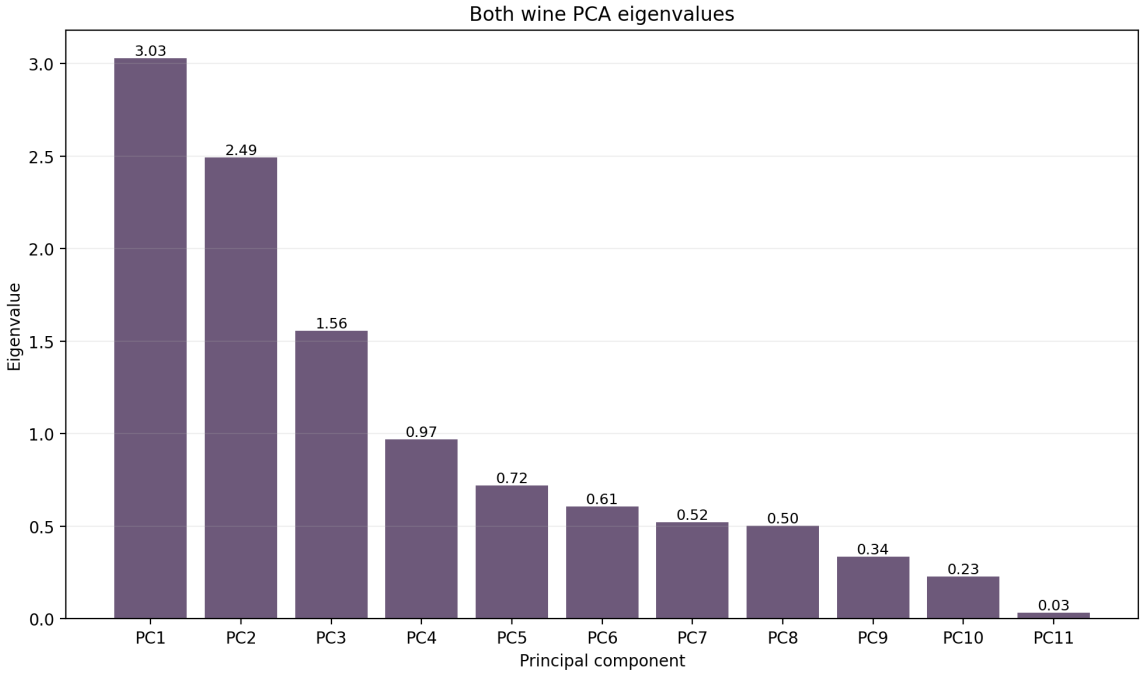

Figure 12: Combined wine PCA eigenvalues